

MOHAMMED ADNAN QURESHI

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SUMMARY

Molecular biologist and discovery team leader with 13+ years' experience in aging biology, assay development, and the translation of preclinical research into actionable consumer health insights. Co-author of high-impact research in Nature Aging, with a proven record in designing, executing, and communicating evidence from preclinical models to support the development of innovative dietary supplements. Skilled in client engagement, cross-functional teamwork, and scientific advocacy, bridging the gap between rigorous research and impactful health solutions.

KEY ACHIEVEMENTS

- Led the development and launch of high-throughput screening platforms supporting the discovery of aging-related interventions.
- Delivered 55+ client projects, translating scientific findings into actionable data and tailored reports for supplement and ingredient innovation.
- Managed SBIR-funded projects advancing novel pro- and postbiotic platforms for metabolic health and aging.
- Collaborated with marketing and regulatory teams to translate research findings into scientifically substantiated health claims for clients.
- Presented research at conferences and engaged with cross-functional partners and external stakeholders to advocate for evidence-based healthy aging solutions.

EXPERIENCE

Apr 2023 - Present

DISCOVERY TEAM LEAD, NemaLife Inc., Lubbock, Texas

- **Project Management & Execution:** Supervised the successful completion of **52+ client projects**, encompassing a range of study designs, readouts, and complexities. Ensured all projects were aligned with goals and delivered on time.
- **Healthy Aging & Product Development:** Supervised and executed projects in aging biology, metabolic health, and dietary supplement innovation. Collaborated with cross-functional teams, including marketing and regulatory, to translate preclinical data into substantiated consumer health claims, and supported product development by generating robust, regulatory-ready data for supplement candidates.
- **Client Engagement & Reporting:** Presented project data to clients through customized reports and presentations, tailoring communications to meet specific client needs. Acted as the primary contact for client inquiries, ensuring timely responses and maintaining strong, collaborative relationships.
- **Leadership & Team Management:** Led lab team and project planning meetings, ensuring clarity in communication regarding timelines, objectives, and responsibilities. Fostered a culture of collaboration and high performance, driving team innovation and productivity.
- **Strategic Resource Planning:** Anticipated staffing requirements and optimized workforce allocation to meet project demands efficiently. Worked across departments to set and achieve KPIs, driving project success.
- **Operational Efficiency & Innovation:** Collaborated with technology development teams to enhance processes, significantly scaling lab operations. Led the development of innovative assays, aligning R&D initiatives with broader business objectives.
- **Regulatory Compliance:** Expertly managed DEA drug inventory, ensuring adherence to regulations and full compliance with DEA Scheduled drug requirements.
- **Grants:** Secured and oversaw **2 Phase 1 SBIR** federal grants as Principal Investigator, managing the project lifecycle and ensuring successful outcomes.

PROJECT SCIENTIST, NemaLife Inc., Lubbock, Texas

- **Assay Development & Innovation:** Spearheaded the development of 13 novel assays for both internal R&D and external clients, enhancing the company's assay portfolio and market competitiveness. Introduced laboratory techniques that improved animal maintenance and bacterial culture, driving productivity.
- **Client-Focused Research & Analysis:** Conducted detailed research on client proposals, designing targeted assays that exceeded expectations. Managed drug preparations for large assays, consistently delivering high-quality samples and data.
- **Laboratory Management & Supervision:** Streamlined laboratory workflows and supervised a team of 4 full-time and 6 part-time researchers, boosting team productivity. Oversaw the maintenance of *C. elegans* and bacterial strains, ensuring robust organism stocks for experiments.
- **Data Analysis & Interpretation:** Utilized GraphPad Prism, R, Excel, and custom scripts for data wrangling, analysis, and interpretation, delivering accurate and actionable insights.
- **Project Management:** Managed project budgets, timelines, and team coordination, ensuring projects remained on track and within scope.

Aug 2016 – Dec 2021

PHD, PELLEGRINO LAB, The University of Texas at Arlington

- Conducted extensive research on the mitochondrial stress response pathway, resulting in the publication of 5 peer-reviewed articles.
- Collaborated with the protein structure lab to successfully express ancient proteins in *C. elegans*, providing valuable insights into the field of genetics.
- Expertly maintained and troubleshooted lab equipment, including microscopes, while also providing training to incoming students.
- Mentored and supervised a team of graduate and undergraduate students, resulting in two students successfully completing their Honors dissertations.
- Taught Anatomy and Physiology to undergraduate nursing and biology majors, receiving positive feedback and exceptional reviews from students.
- Organized and led workshops on Software Carpentry, aimed at introducing scientists to computational approaches in research.
- Utilized graphic design skills to design graphics for lab scientific papers
- Actively recruited incoming graduates to the lab.

SCIENTIFIC SKILLS

Molecular Biology: PCR, qPCR, Molecular Cloning, Protein expression and purification, DNA and Protein gels, Western Blot, Protein-protein interaction Assay, DNA and RNA extraction, Plasmid purification, Chromatography (Size Exclusion, Affinity, Reverse Phase, Ion Exchange)

Microscopy: Fluorescence, Confocal and Bright-field imaging (ImageXpress Micro XLS, ZEISS SteREO Discovery.V20, ZEISS Axio Imager 2 and Stemi 305)

Microbiology: Bacterial culture, bacterial growth curve, SLAY (antimicrobial peptide analysis), Transformation, Bacterial fluorescence tagging and imaging

Animal studies:

***Caenorhabditis elegans*:** Maintenance, RNAi screening, Drug screening, Genetic crosses, EMS mutagenesis screening, Lifespan assay, Survival assay, extrachromosomal DNA expression

***Mus musculus*:** Dissection, Drug administration, Bone marrow cell extraction

Cell Biology: Cell culture, Cell-viability assay, Transfection

Computational Skills: R, RNA-Seq Analysis, Linux, GraphPad Prism, ImageJ, Zeiss Zen, MetaXpress, Adobe Illustrator

SBIR GRANTS (PI):

1R43AT012449-01A1 - Development of a high throughput in vivo screening platform for identifying novel pro-and postbiotic bacterial strains

1R43AA031342-01A1 - Development of an AI-assisted therapeutic discovery platform for Alcohol Use Disorders using an invertebrate model

EDUCATION

Aug 2016 - Present

PHD - QUANTITATIVE BIOLOGY, The University of Texas at Arlington

May 2013

MASTER OF SCIENCE - BIOTECHNOLOGY, Vellore Inst. Of Technology

Mar 2011

BACHELOR OF SCIENCE - BIOTECHNOLOGY, University of Mumbai

PUBLICATIONS

Civiletto, G., Brunetti, D., Lizzo, G., ... **Qureshi, M. A.**, ... & Gut, P. (2025). Herbal terpenoids activate autophagy and mitophagy through modulation of bioenergetics and protect from metabolic stress, sarcopenia and epigenetic aging. **Nature Aging.**

Mahmud, S. A., **Qureshi, M. A.**, & Pellegrino, M. W. (2021). On the offense and defense: mitochondrial recovery programs amidst targeted pathogenic assault. *The FEBS Journal*

Sapkota, M., **Qureshi, M. A.**, Balikosa, Y., Nguyen, C., Boll, J.M., & Pellegrino, M. W. (2021). A nematode-derived, mitochondrial stress signaling-regulated peptide exhibits broad antibacterial activity. *Biology Open.*

Mahmud, S. A., **Qureshi, M. A.**, Sapkota, M., & Pellegrino, M. W. (2020). A pathogen branched-chain amino acid catabolic pathway subverts host survival by impairing energy metabolism and the mitochondrial UPR. *PLoS Pathogens*. **[co-first author]**

Qureshi, M. A., Haynes, C. M., & Pellegrino, M. W. (2017). The mitochondrial unfolded protein response: Signaling from the powerhouse. *Journal of Biological Chemistry*.